

2026 양자정보경진대회 지정문제 1번

Classification

Trotterized Quantum Simulation

Background

Trotterization is one of the most widely used approaches to implementing Hamiltonian time evolution on a quantum computer. By decomposing the exponential of a Hamiltonian into a product of simpler exponentials, this practical method for quantum simulation exhibits a trade-off between approximation accuracy and circuit depth.

Although this asymptotic trade-off is suboptimal compared with quantum singular value transformation, its simplicity at small scales makes it suitable for applications in the early fault-tolerant regime. In this challenge, participants will study Trotterized time evolution for a periodic Heisenberg model, analyze the resulting circuit structure, and investigate how spectral information can be extracted from the dynamics. Teams are encouraged to build on the official quantum simulation materials, replacing the tutorial Hamiltonian with the Heisenberg model below and exploring it creatively. The project may include simulation, circuit optimization, controlled time evolution, or spectral estimation based on autocorrelation signals.